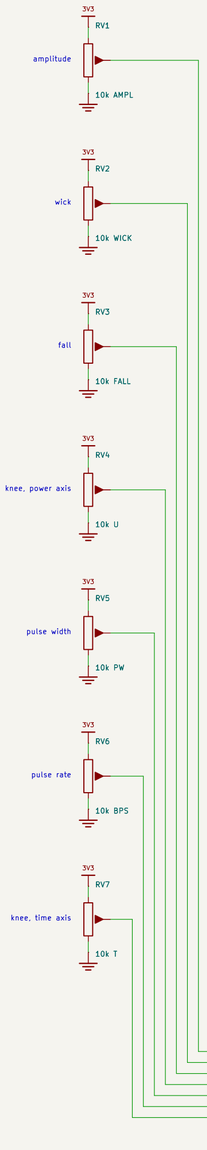
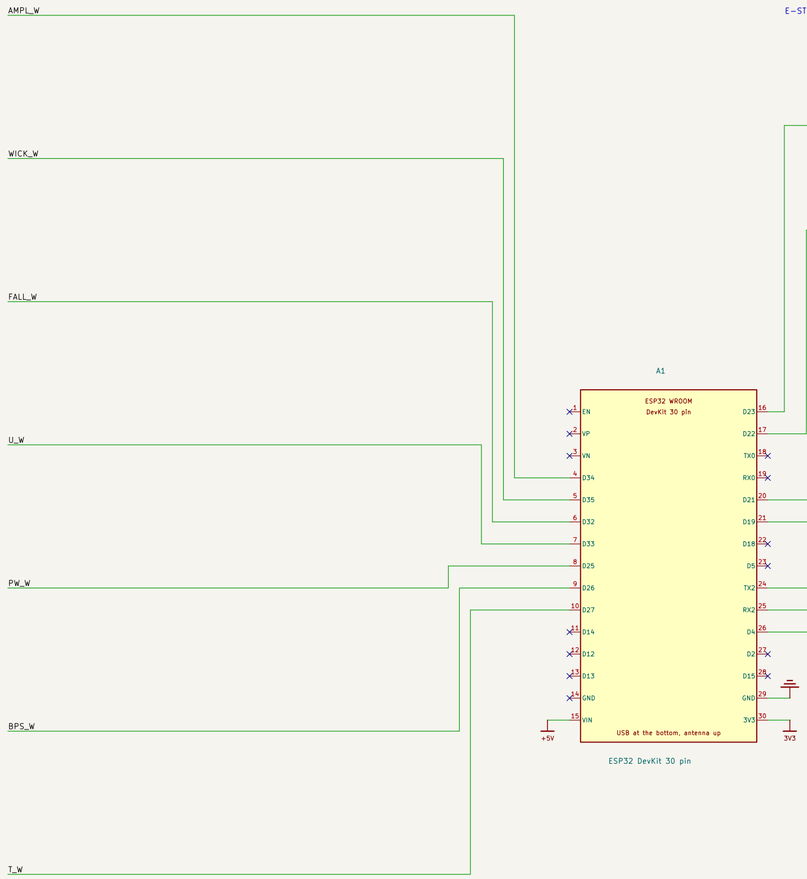
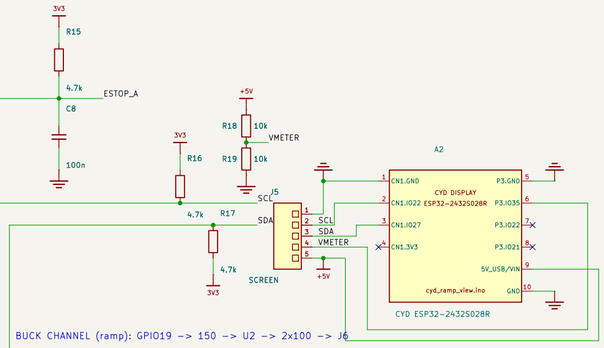
QCW DRSSTC CONTROLLER v5.9-ESP32 — schematic for the control board (rev. E, 2026-09-27): ESP32 DevKit in socket headers, 2x TC4420 DIP-8, through-hole resistors, SMD 1206 capacitors (on the copper side), knobs daisy-chained on J1, everything else on J2...J9.





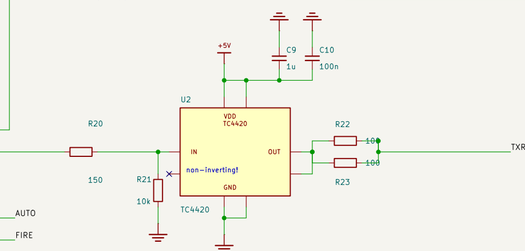
E-STOP group A: GPIO23 -> label ESTOP_A -> J4.1; the switch shorts it to J4.2 = GND. 4.7k to 3V3 and 100n at the pin.



DISPLAY is off-board, 5-wire cable to J5:
1 GND -> CN1.GND, 2 SCL -> CN1.IO22, 3 SDA -> CN1.IO27,
4 VMETER -> P3.IO35, 5 +5V -> display USB/VIN.
4.7k pull-ups and the 10k/10k divider are on the board.
CN1.3V3 is not used.

VOLTMETER: R18 from +5V to node VMETER, R19 from the node to GND;
the display shows e.g. 5.0V in small font: green >4.8 V, yellow, red <4.5 V.

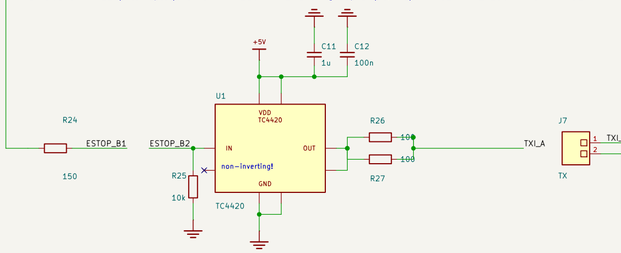
BUCK CHANNEL (ramp): GPIO19 -> 150 -> U2 -> 2x100 -> J6



HFBR-1412/1414 transmitter is off-board, 2-wire cable to J: 1 = anode (pins 2, 6, 7 tied together).
2 = cathode (pin 3). Pins 1/4/5/8 unused. It needs no supply: ~60 mA comes from the TC4420 through 2x100 ohm.

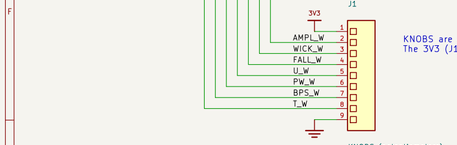
AUTO (TX2=GPIO17), FIRE (RX2=GPIO16): labels go to J3/J2.

BRIDGE CHANNEL (interrupter): GPIO4 -> 150 -> E-STOP group B (J8, NC) -> U1 -> 2x100 -> J7 -> D1



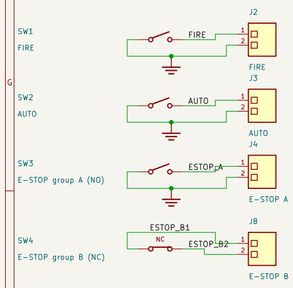
DRIVER BOARD (qcw_driver)
RX buck HFBR-2412 <- fiber from D2
RX bridge HFBR-2412 <- fiber from D1
Receiver: dark = HIGH.
Jumper J1: position C-B
(light = ON, broken fiber = OFF).
Receivers: 2 = Vcc +5 V of that board,
3 and 7 = common, 6 = output.
NO galvanic connection to the controller.

HFBR-1412/1414 transmitter is off-board, 2-wire cable to J: 1 = anode (pins 2, 6, 7 tied together).
2 = cathode (pin 3). Pins 1/4/5/8 unused. It needs no supply: ~60 mA comes from the TC4420 through 2x100 ohm.

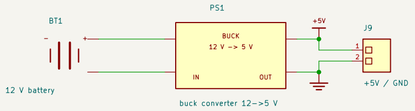


KNOBS are off-board, NO extra resistors: each wiper goes straight to its ESP32 pin through J1.2...J1.8 (one wire each).
The 3V3 (J1.1) and GND (J1.9) rails are daisy-chained from pot to pot — one pair of wires from the board, 3.3 V across every pot.

Light = bridge enabled; broken fiber = dark = inhibit. On the driver board set J1 to position C-B (light = ON).



BUTTONS are off-board: FIRE and AUTO on J2/J3 (signal, GND), pull-ups inside the ESP32.
E-STOP is one switch with two contact groups, two 2-wire cables: J4 (group A, NO) shorts GPIO23 to ground; J8 (group B, NC) is in series with the U1 input.



POWER: battery -> off-board buck converter -> screw terminal J9 (1 = +5 V, 2 = GND). On the board +5 V feeds ESP32 VIN, U1/U2 VDD, R18 and J5.5 (display).
3V3 comes from the ESP32 module (to J1...J7 and the pull-ups). Ground: buck minus = J9.2; the whole board ground is the bottom-layer pour. Enclosure to J9.2 with a single wire.
Off-board (on_board=no): RV1...RV7, SW1...SW4, A2 display, D1/D2 transmitters, PS1, BT1. Labels with the same name = one net (cable <-> board).

Author: Mikola_Tesla — License: CC BY 4.0
ESP32 DevKit + 2x TC4420 DIP: knobs, display, fiber TX, buck on connectors

Sheet: /
File: qcw_board.kicad_sch

Title: QCW DRSSTC controller: control board schematic

Size: A2	Date: 2026-09-27	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 1/1